Requirements Gathering

Stakeholders, User Stories, and System Requirements

Podcast Streaming Platform

Graduation Project - Faculty of Computers and Information

Backend: ASP.NET Core Web API | Frontend: React / Next.js | Database: PostgreSQL

Prepared by: Project Development Team

Academic Year 2025 - 2026

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1. Introduction

This document defines all requirements for the Podcast Streaming Platform, a web-based application for discovering, streaming, and managing podcast content extracted from YouTube videos. Requirements were gathered through stakeholder interviews, market research, and competitive analysis of platforms like Spotify and Apple Podcasts. Each requirement is categorized, prioritized (MoSCoW method), and traceable to its originating stakeholder. The platform targets three user roles: Listeners, Content Creators, and Administrators, built with ASP.NET Core 8, React/Next.js, and PostgreSQL 16 over a 16-week Agile Scrum timeline.

2. Stakeholder Analysis

2.1 Stakeholder Register

Stakeholder	Category	Role/Interest	Key Concerns
End Users (Listeners)	Primary	Browse, search, stream, organize podcasts	Audio quality, ease of use, fast loading
Content Creators	Primary	Submit YouTube URLs, trim audio, manage uploads	Easy upload, trimming precision, analytics
Platform Admins	Primary	Manage users, content, monitor activity	Admin tools, analytics, content moderation
Development Team	Secondary	Design, build, test, deploy the platform	Clear requirements, feasible timeline
University Supervisors	Secondary	Academic guidance and evaluation	Documentation quality, code standards
YouTube API (yt-dlp)	Secondary	Audio extraction service	API stability, rate limits, terms of service
Cloud Providers	Secondary	Hosting, storage, CDN	Uptime, cost, scalability

Table 2.1: Stakeholder register with roles and concerns.

2.2 Influence-Interest Matrix

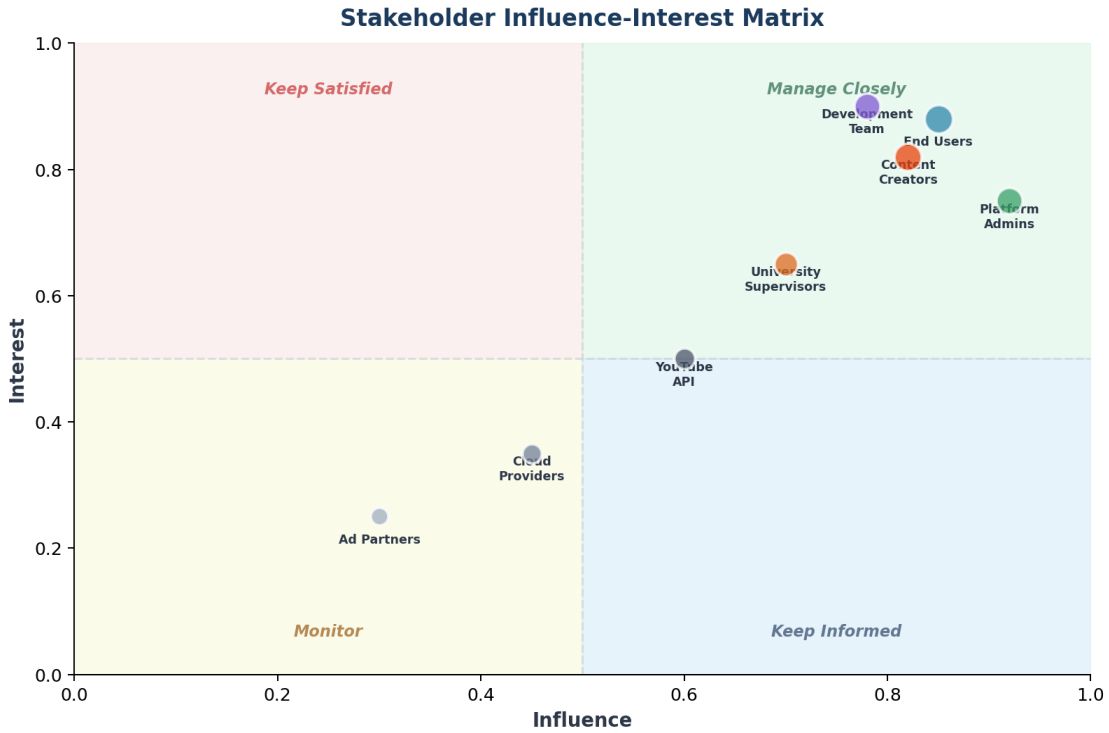

Figure 2.1: Stakeholder Influence-Interest Matrix with engagement strategy quadrants.

2.3 Communication Strategy

Stakeholder	Quadrant	Frequency	Channel
End Users	Manage Closely	Continuous	In-app surveys, beta testing, feedback forms
Content Creators	Manage Closely	Weekly	Sprint demos, feature previews, direct email
Platform Admins	Manage Closely	Daily/Weekly	Stand-ups, admin portal, Slack channel
Development Team	Manage Closely	Daily	Stand-ups, Jira board, code reviews, Git
University Supervisors	Keep Satisfied	Bi-weekly	Progress reports, presentations, docs
YouTube API	Keep Satisfied	As needed	API documentation, version monitoring

Stakeholder	Quadrant	Frequency	Channel
Cloud Providers	Keep Informed	Monthly	Status dashboards, SLA reports

Table 2.2: Communication plan by stakeholder group.

3. User Stories & Use Cases

3.1 User Personas

Persona	Description
Ahmed - Casual Listener	24, marketing associate, commutes 45 min. Wants audio-only podcast experience without YouTube battery drain. Needs playlists, fast search, smooth streaming. Tech: Intermediate.
Sara - Content Creator	28, software engineer, weekly tech podcast host. Wants to convert 90+ min YouTube videos into bite-sized audio clips. Needs precise trimming, batch processing, analytics. Tech: Advanced.
Mohamed - Admin	30, IT professional, maintains platform health. Needs real-time dashboard, content moderation tools, user management, and system monitoring. Tech: Expert.

Table 3.1: User personas representing target audience segments.

3.2 User Stories

ID	User Story	Priority	Acceptance Criteria
US-01	Listener wants to register so they can access personalized features	Must	Account created; verification sent; login < 3s
US-02	Listener wants to browse trending podcasts on home page	Must	Home page loads < 2s; shows 12+ items
US-03	Listener wants to search by title/topic/creator	Must	Results < 1s; partial text matching
US-04	Listener wants play/pause/skip with web player	Must	Controls respond < 200ms; seeking supported
US-05	Listener wants to create/manage playlists	Should	Add/remove/reorder; persists across sessions
US-06	Listener wants to favorite episodes for quick access	Should	Toggle works instantly; favorites page available
US-07	Creator wants to paste YouTube URL to extract audio	Must	URL validated < 2s; extraction starts automatically
US-08	Creator wants to trim audio with start/end times	Must	Quality preserved; preview of trimmed segment
US-09	Creator wants to add metadata (title, tags, description)	Should	Metadata saved and shown in search results
US-10	Admin wants dashboard with platform metrics	Must	Dashboard loads < 3s; real-time stats
US-11	Admin wants to manage user accounts (activate/deactivate)	Must	Changes immediate; audit trail maintained
US-12	Admin wants content reports and analytics	Should	Reports < 5s; date range filtering

Table 3.2: User stories with priorities and acceptance criteria.

3.3 Use Case Diagram

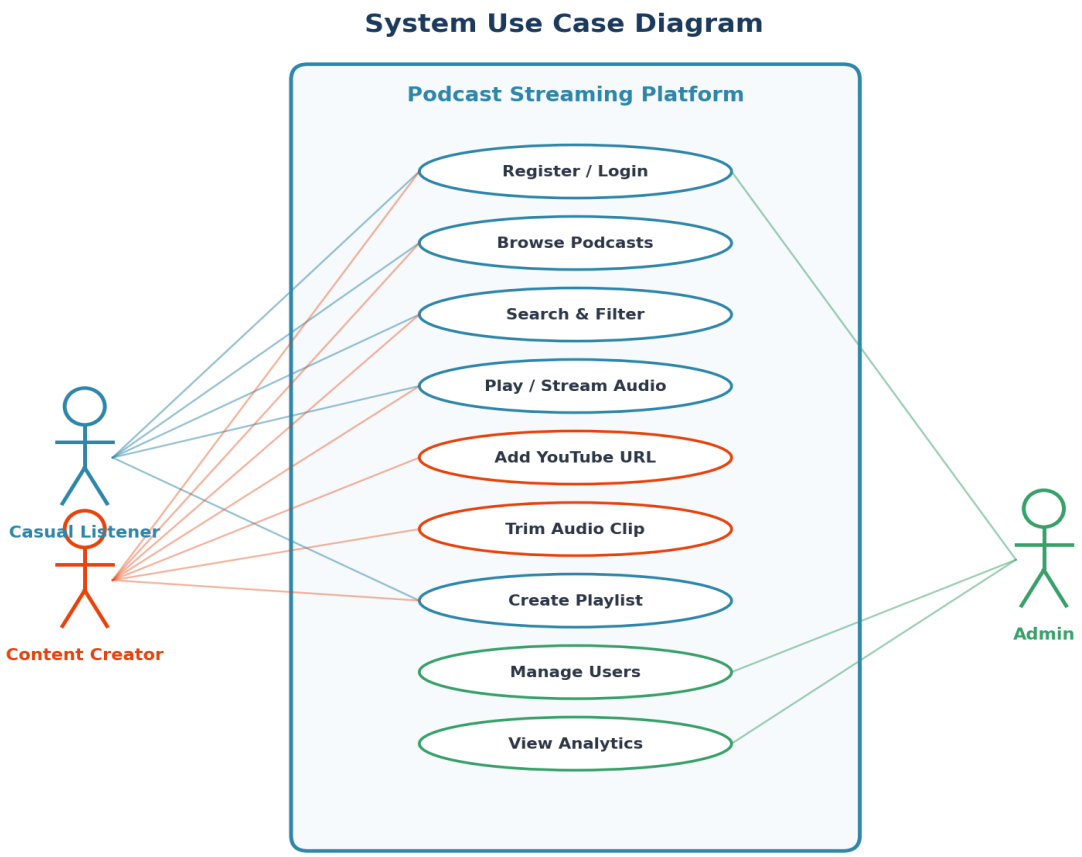

Figure 3.1: System use case diagram with three primary actors.

3.4 Detailed Use Cases

UC-01: Extract Audio from YouTube URL

Property	Description
Actor	Content Creator
Main Flow	1. Navigate to Add Podcast page 2. Paste YouTube URL 3. System validates URL 4. Display metadata preview 5. Confirm extraction 6. yt-dlp downloads audio 7. FFmpeg normalizes format 8. Store audio + create record
Alternative Flows	3a. Invalid URL: show error 6a. Video unavailable: show error 6b. Timeout: retry once, then notify
Postconditions	Audio/record stored; user redirected to management

Table 3.3: UC-01 detailed description.

UC-02: Trim Audio Clip

Property	Description
Actor	Content Creator
Main Flow	1. Open audio editor 2. View waveform visualization 3. Set start time 4. Set end time 5. Preview segment 6. Click Trim & Save 7. FFmpeg cuts audio 8. Replace original file
Alternative Flows	3a. Start > End: auto-swap 5a. Preview error: retry 7a. Trim fails: log error
Postconditions	Audio/record stored; user redirected to management

Table 3.4: UC-02 detailed description.

UC-03: Search & Discover Podcasts

Property	Description
Actor	Casual Listener
Main Flow	1. Navigate to search bar 2. Enter query 3. Full-text search across metadata 4. Display ranked results 5. Apply filters (category, duration) 6. Click episode for details
Alternative Flows	2a. Empty query: show trending 4a. No results: show suggestions
Postconditions	Audio/record stored; user redirected to management

Table 3.5: UC-03 detailed description.

UC-04: View Analytics Dashboard

Property	Description
Actor	Platform Admin
Main Flow	1. Navigate to Dashboard 2. View key metrics (users, plays, podcasts) 3. View trend charts 4. Select date range 5. Export report (CSV/PDF)
Alternative Flows	2a. Insufficient data: placeholder 6a. Export fails: show notification
Postconditions	Audio/record stored; user redirected to management

Table 3.6: UC-04 detailed description.

4. Functional Requirements

62 functional requirements across 9 modules. Each has a unique ID (FR-XXX), priority (MoSCoW), and acceptance criteria.

4.1 User Management

Req ID	Requirement	Priority	Acceptance Criteria
FR-101	User Registration (email+password)	Must	Email validated; min 8 char password; verification email sent
FR-102	Email Verification	Must	Link expires 24h; account activated after verification
FR-103	JWT Authentication	Must	Access token 15min + refresh 7d; RS256 signing
FR-104	Password Reset	Must	Reset link via email; 1h expiry; complexity enforced
FR-106	Role-Based Access Control	Must	3 roles: Listener, Creator, Admin; per-endpoint enforcement
FR-105	Profile Management	Should	Update name, avatar, bio; reflected immediately
FR-107	Session Management	Should	Max 3 concurrent; invalidated on password change

4.2 Podcast Content Management

Req ID	Requirement	Priority	Acceptance Criteria
FR-201	Submit YouTube URL	Must	URL validated; metadata fetched; extraction initiated < 5s
FR-204	Categorize podcasts	Must	Predefined categories; multi-select; filterable in search
FR-205	Content pagination	Must	20 items/page; sortable by date, popularity, title
FR-206	Content detail page	Must	Title, description, creator, duration, play count, related
FR-207	Content moderation	Must	Admin can flag, approve, remove; actions logged
FR-203	Edit metadata	Should	Title, description, tags form; instant save

4.3 Audio Processing

Req ID	Requirement	Priority	Acceptance Criteria
FR-301	YouTube audio extraction	Must	MP3/M4A format; 44.1kHz+; 128kbps+ bitrate
FR-302	Format normalization	Must	Consistent output format; metadata tags embedded
FR-303	Audio trimming	Must	Millisecond accuracy; no artifacts at cut points
FR-305	Background job queue	Must	Async processing; status: pending/processing/done/failed
FR-307	Error handling	Must	Retry up to 2x; error details logged and displayed

4.4 Streaming & Playback

Req ID	Requirement	Priority	Acceptance Criteria
FR-401	Web audio player	Must	Play, pause, skip, volume; responds < 200ms
FR-402	Progress bar with seek	Must	Click-to-seek; current/total time display
FR-404	Progressive streaming	Must	Audio plays within 2s; buffering indicator shown
FR-403	Playback speed	Should	0.5x-2x in 0.25 increments; persists per user

4.5 Search & Discovery

Req ID	Requirement	Priority	Acceptance Criteria
FR-501	Full-text search	Must	Title, description, tags, creator; relevance ranked
FR-502	Partial/fuzzy matching	Must	Partial words + typo tolerance; < 500ms
FR-503	Filter by category/duration/date	Must	Multi-filter combinable; instant results
FR-505	Trending content	Should	Updated daily; based on popularity

4.6 Playlists & Favorites

Req ID	Requirement	Priority	Acceptance Criteria
FR-601	Create/edit/delete playlists	Must	Name required; optional description and cover
FR-602	Add/remove episodes	Must	Add via button; remove in management view
FR-604	Bookmark favorites	Should	One-click toggle; count displayed

4.7 Notifications

Req ID	Requirement	Priority	Acceptance Criteria
FR-701	In-app notification center	Must	Bell icon + unread count; mark as read
FR-703	Job completion notification	Must	Success/failure distinction; link to content

4.8 Analytics & Reporting

Req ID	Requirement	Priority	Acceptance Criteria
FR-801	Admin dashboard metrics	Must	Total users, active, podcasts, plays; refresh 5min
FR-804	Export reports	Should	CSV data / PDF visual; date range filter

4.9 Admin Dashboard

Req ID	Requirement	Priority	Acceptance Criteria
FR-901	User account management	Must	View, activate, deactivate, delete; search by email
FR-902	Role assignment	Must	Immediate effect; audit log updated
FR-903	Moderation queue	Must	Reported content; approve/reject/remove with reason

4.10 Functional Requirements Summary

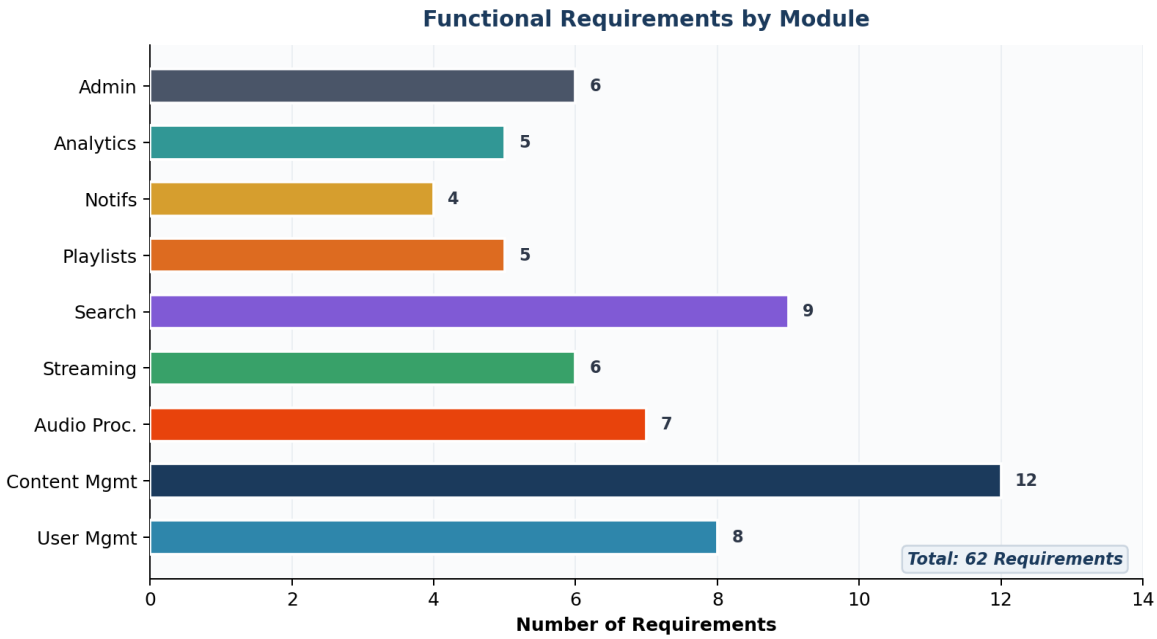

Figure 4.1: Functional requirements distribution by module.

Module	Must	Should	Could	Total
User Management	5	3	0	8
Content Management	5	4	3	12
Audio Processing	5	2	0	7
Streaming/Playback	4	1	1	6
Search/Discovery	4	2	3	9
Playlists/Favorites	2	2	1	5
Notifications	2	1	1	4
Analytics	1	3	1	5
Admin Dashboard	3	2	1	6
Total	31	20	11	62

Table 4.2: Functional requirements summary by module and priority.

5. Non-Functional Requirements

5.1 Performance

Req ID	Description
NFR-101	Page load < 2s (normal load)
NFR-102	API response < 500ms (p95)
NFR-103	Search < 1s
NFR-104	Audio buffer < 2s
NFR-105	500 concurrent users min
NFR-106	Audio extraction < 60s for 30min video
NFR-107	DB query < 200ms (indexed)

5.2 Security

Req ID	Description
NFR-201	bcrypt/Argon2 password hashing
NFR-202	JWT (15min access + 7d refresh, RS256)
NFR-203	HTTPS (TLS 1.2+) with HSTS
NFR-204	Parameterized queries + CSP for XSS
NFR-205	Rate limit: 100/min standard, 20/min auth
NFR-206	Strict CORS (no wildcard in prod)
NFR-207	Client + server input validation
NFR-208	Audit log for security events

5.3 Usability

Req ID	Description
NFR-301	Responsive: 320px-2560px
NFR-302	WCAG 2.1 AA: contrast 4.5:1, keyboard nav
NFR-303	Max 3 clicks to any core feature
NFR-304	Onboarding tour for new users
NFR-305	Clear, actionable error messages
NFR-306	Consistent UI via shared component library

5.4 Reliability

Req ID	Description
NFR-401	99.5% uptime (business hours)
NFR-402	Daily DB backups, 30-day retention
NFR-403	Audio files replicated (2+ locations)
NFR-404	Graceful error handling, no unhandled exceptions
NFR-405	Retry with exponential backoff (max 3)
NFR-406	FK constraints, transactions, checksums

5.5 Scalability

Req ID	Description
NFR-501	Stateless API for horizontal scaling
NFR-502	Redis cache for trending, sessions, search
NFR-503	CDN for audio delivery
NFR-504	DB connection pooling
NFR-505	Async processing for heavy tasks

5.6 Compatibility

Req ID	Description
NFR-601	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
NFR-602	iOS Safari 14+, Chrome Android 90+
NFR-603	320px to 2560px screen support
NFR-604	MP3, AAC, OGG, WAV format support
NFR-605	.NET 8+ on Linux (Ubuntu 22.04+)

5.7 Maintainability

Req ID	Description
NFR-701	Clean Architecture (Projects: API, Core, Infrastructure)
NFR-702	XML doc comments + Swagger API docs
NFR-703	Unit test coverage >= 60%
NFR-704	Docker containerization + docker-compose
NFR-705	CI/CD pipeline (GitHub Actions)
NFR-706	Environment-based config (Dev/Staging/Prod)
NFR-707	URL-based API versioning (/api/v1/)

5.8 NFP Priority Assessment

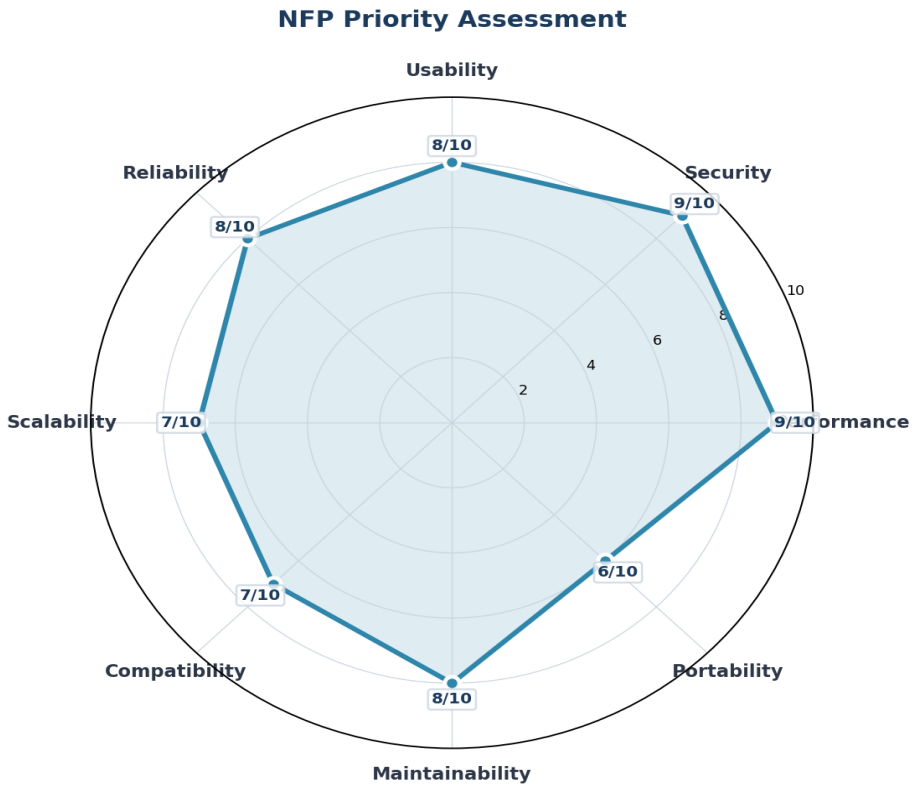

Figure 5.1: Non-functional requirements priority radar (1-10 scale).

6. Requirements Prioritization

6.1 MoSCoW Prioritization

Must Have (50%): Core MVP features - auth, audio extraction/playback, content browsing, search, basic admin. Should Have (30%): Important enhancements - playback speed, moderation, analytics. Could Have (20%): Nice-to-have - scheduling, auto-play, advanced search. Won't Have: Native mobile apps, monetization, RSS import.

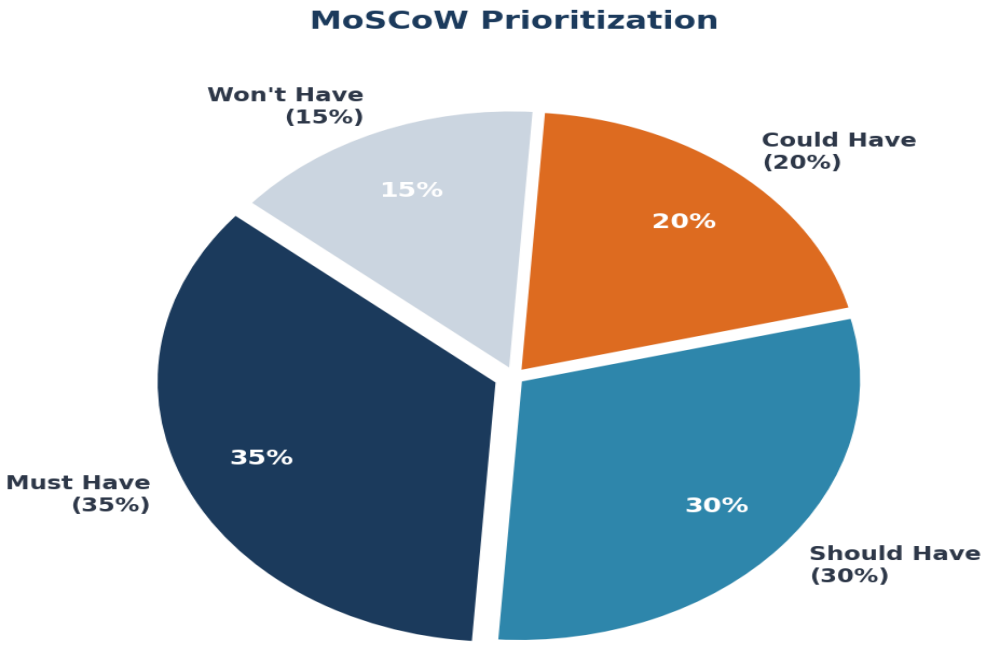

Figure 6.1: MoSCoW requirements distribution.

6.2 Prioritization Matrix

Priority	Module	Requirements	Sprint	Effort
Must	User Management	FR-101-104, FR-106	Sprint 1-2	3 weeks
Must	Content Management	FR-201,202,204-206	Sprint 2-4	4 weeks
Must	Audio Processing	FR-301-303,305,307	Sprint 3-5	5 weeks
Must	Streaming/Playback	FR-401,402,404	Sprint 4-5	2 weeks
Must	Search/Discovery	FR-501-504	Sprint 5-6	2 weeks
Must	Notifications	FR-701,703	Sprint 6-7	1 week
Must	Admin Dashboard	FR-901-903	Sprint 7-8	2 weeks
Should	All Modules	FR-105,107,108,203,207,208,304,306,403,405,505,506,601-604,702,801-804,904-905	Sprint 6-10	10 weeks
Could	All Modules	FR-209,210,212,406,507-509,605,704,805,906	Sprint 10-11	5 weeks

Table 6.1: Requirements prioritization matrix with sprint allocation.

7. Requirements Traceability

7.1 Traceability Matrix

Req ID	Requirement	Stakeholder	User Story	Use Case
FR-101	User Registration	Listener, Creator	US-01	UC-01
FR-103	JWT Authentication	All Users	US-01	UC-01
FR-201	YouTube URL Submission	Creator	US-07	UC-01
FR-301	Audio Extraction	Creator	US-07	UC-01
FR-303	Audio Trimming	Creator	US-08	UC-02
FR-401	Web Audio Player	Listener	US-04	UC-03
FR-501	Full-text Search	Listener	US-03	UC-03
FR-601	Playlist Management	Listener	US-05	-
FR-701	Notification Center	All Users	US-07	-
FR-801	Admin Metrics	Admin	US-10	UC-04
FR-901	User Account Mgmt	Admin	US-11	UC-04
FR-903	Content Moderation	Admin	US-10	UC-04

Table 7.1: Bidirectional traceability linking requirements to stakeholders and use cases.

7.2 Validation & Sign-off

Reviewer	Name	Date	Status
Project Manager	[TBA]	-	Pending
Technical Lead	[TBA]	-	Pending
University Supervisor	[TBA]	-	Pending
QA Lead	[TBA]	-	Pending
Product Owner	[TBA]	-	Pending

Table 7.2: Requirements document sign-off record.

Version	Date	Author	Changes
1.0	2026-04-30	Project Team	Initial condensed release

Table 7.3: Document revision history.